

Multiple Choice

1. **Answer:** B

Options: A) 1.6 ns; B) 2.5 ns; C) 4.2 ns; D) 6.9 ns

Note: The correct answer is 2.5 ns because, in the observer's reference frame, the length contraction of the meter stick due to its speed of $0.8c$ results in an effective length of approximately 0.6 meters, and the time taken to pass the observer is calculated using the formula $\text{time} = \text{length}/\text{speed}$, which gives $0.6 \text{ m} / (0.8c) = 2.5 \text{ ns}$.

2. **Answer:** B

Options: A) 2; B) 250; C) 5,000; D) 10,000

Note: The resolving power of a spectrometer is defined as the ratio of the wavelength to the minimum difference in wavelength that can be resolved, and since the difference between the two wavelengths is 2 nm (502 nm - 500 nm), the resolving power is calculated as $500 \text{ nm} / 2 \text{ nm} = 250$.

3. **Answer:** D

Options: A) Specific heat; B) Thermal conductivity; C) Electrical resistivity; D) Hall coefficient

Note: The Hall coefficient indicates the type of charge carriers in a doped semiconductor-positive or negative-by analyzing the direction of the induced voltage resulting from a magnetic field applied perpendicular to the current flow.

4. **Answer:** C

Options: A) $1.2 \times 10^9 \text{ m/s}$; B) $3.0 \times 10^8 \text{ m/s}$; C) $1.5 \times 10^8 \text{ m/s}$; D) $1.0 \times 10^8 \text{ m/s}$

Note: The speed of light in a dielectric medium is calculated using the formula c/n , where c is the speed of light in a vacuum (approximately $3.0 \times 10^8 \text{ m/s}$) and n is the refractive index, which is the square root of the dielectric constant; thus, for a dielectric constant of 4.0, the refractive index n is 2.0, resulting in a speed of light of $1.5 \times 10^8 \text{ m/s}$.

5. **Answer:** A

Options: A) Electron; B) Meson; C) Photon; D) Boson

Note: The negative muon (μ^-) is a leptonic particle that shares similar properties with the electron, such as

True / False

6. **Answer:** True

Options: A) True; B) False

Note: Electromagnetic radiation emitted from a nucleus is most likely in the form of gamma rays because gamma rays are high-energy photons released during nuclear

decay processes when the nucleus transitions from a higher energy state to a lower energy state.

7. **Answer:** True

Options: A) True; B) False

Note: The statement is true because the core processes of the Sun involve nuclear fusion, where four hydrogen nuclei combine to form a helium nucleus, releasing energy as a result of the mass defect, which is described by Einstein's equation $E=mc^2$.

8. **Answer:** True

Options: A) True; B) False

Note: The resolving power of a grating spectrometer is defined as the ratio of the wavelength to the minimum difference in wavelength that can be resolved, and for wavelengths of 500 nm and 502 nm, the resolving power is indeed 250, which allows for distinguishing between these two closely spaced wavelengths.

9. **Answer:** True

Options: A) True; B) False

Note: A dye laser is considered the best type of laser for spectroscopy across a range of visible wavelengths because it can be tuned to emit a wide variety of wavelengths by changing the dye solution, allowing it to match specific absorption or emission lines of different molecules effectively.

10. **Answer:** False

Options: A) True; B) False

Note: The statement is false because for a given azimuthal quantum number l , the magnetic quantum number m_l can take on values ranging from $-l$ to $+l$, which for $l = 2$ allows for a total of 5 possible values: -2 , -1 , 0 , $+1$, and $+2$.

Long Form Response

11. **Answer:** In the hydrogen atom, an electron in the $n = 4$, $l = 1$ state can transition to a variety of lower energy states, specifically those allowed by the selection rules of quantum mechanics. The primary selection rules dictate that the change in the principal quantum number (n) can be any integer, while the change in the azimuthal quantum number (l) must be ± 1 . Therefore, an electron in the $n = 4$, $l = 1$ state can transition to the $n = 3$, $l = 0$ state or the $n = 3$, $l = 2$ state. However, transitions to states such as $n = 4$, $l = 1$ or $n = 2$, $l = 1$ are not allowed, as they violate the selection rules regarding changes in l or do not result in a decrease in energy. Thus, the only reachable final state from $n = 4$, $l = 1$ is $n = 3$, $l = 1$, which adheres to these quantum mechanical principles.

Note: correct because it accurately applies the quantum mechanical selection rules, specifically that l must be ± 1 , allowing transitions from the $n = 4$, $l = 1$ state to the $n = 3$, $l = 0$ and $n = 3$, $l = 2$ states, while also recognizing that certain final states are not reachable due to these rules.

12. **Answer:** In a one-dimensional inelastic collision between a particle of mass $2m$ and a stationary mass m , the initial kinetic energy is solely due to the moving mass, given by $KE_{\text{initial}} = \frac{1}{2} (2m)v^2 = mv^2$, where v is the initial velocity of the moving mass. After the collision, the two masses stick together and move with a common velocity, which can be found using conservation of momentum: $2mv = (2m + m)V$, leading to $V = \frac{2}{3}v$. The final kinetic energy of the combined mass is $KE_{\text{final}} = \frac{1}{2} (3m)(\frac{2}{3}v)^2 = \frac{2}{3}mv^2$. The fraction of kinetic energy lost during the collision is then calculated as $(KE_{\text{initial}} - KE_{\text{final}}) / KE_{\text{initial}} = (mv^2 - \frac{2}{3}mv^2) / mv^2 = \frac{1}{3}$. Thus, the collision results in a loss of one-third of the initial kinetic energy.
- Note:* correct because it correctly applies the principles of conservation of momentum and kinetic energy to determine the initial kinetic energy, the final velocity after the inelastic collision, and subsequently calculates the fraction of kinetic energy lost during the collision.

End of Answer Key